

UTKARSH PRAKASH

Data Structures | Backend Developer | Problem Solver

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PROFESSIONAL SUMMARY

Backend-focused Computer Science undergraduate experienced in building REST APIs, payment-integrated systems, and asynchronous backend services. Comfortable working across relational data modeling, caching, and background job processing, with strong data structures and algorithms fundamentals applied to building reliable, production-oriented applications.

EDUCATION

Babu Banarasi Das Institute of Technology & Management | Lucknow
B.Tech CSE

Oct 2023
CGPA : 8.2

Bradford International School | Patna
12th

Apr 2022
Percentage : 80 %

Bradford International School | Patna
10th

Apr 2020
Percentage : 87 %

PROJECTS

Astra | △ Vercel ☞ Document

Jun 2026 - Jul 2026

- Built an end-to-end appointment booking and payment workflow — slot reservation, Razorpay payment verification, booking confirmation, and failure/expiry handling — to keep bookings consistent under concurrent requests.
- Developed modular **REST APIs** for authentication, bookings, payments, and consultations in **Node.js/Express**, utilizing **PostgreSQL** accessed through **Prisma ORM** for type-safe queries and migrations.
- Implemented real-time video consultations using **WebRTC** and [Socket.IO](#), alongside backend-triggered automated report generation and notifications.

Technologies : React, Node.js, Express, MongoDB, Tailwind CSS, REST APIs, WebRTC, Socket.IO

Placement Portal

Jun 2026 - Jul 2026

Full-Stack Developer

- Designed scalable **REST APIs** in Flask for a multi-role platform (students, recruiters, administrators) covering job postings, applications, and eligibility workflows, backed by a **PostgreSQL schema** managed with SQLAlchemy.
- Integrated **Redis** for caching and **Celery** for asynchronous background job processing, reducing load on synchronous request paths and improving responsiveness for resource-intensive operations.

Technologies : Flask, Python, SQLAlchemy, Vue.js, Bootstrap, Redis, Celery, REST APIs, PostgreSQL, web scraping, data extraction

WebAiBuilder

- Built the backend for an **AI-powered website generator** that transforms natural-language prompts into React project scaffolds using the OpenRouter API, including an iterative update pipeline for AI-driven code changes.
- Engineered supporting API endpoints for **live preview**, manual code editing, and project export/publishing.

Technologies : React, Node.js, Express.js, MongoDB, OpenRouter, JavaScript, REST APIs

SKILLS

Programming Languages :

C++, Python, JavaScript

Backend :

Express.js, Flask, REST APIs

Frontend :

React.js, Next.js, Vue.js, Bootstrap, Tailwind CSS

Databases :

PostgreSQL, MySQL, SQLite, MongoDB, Redis

AI/ML :

LLM Integration, AI Agent Workflows, NLP, Semantic Search, Embeddings, Vector Search, RAG, Machine Learning, Natural Language Processing, LangChain

Core Concepts :

OOPs, DBMS, Operating Systems, Computer Networks, System Design, DSA

Tools & Platforms :

Git, GitHub, Postman, Docker, WebRTC, Socket.IO, Vercel, Render, CI/CD

ACHIEVEMENTS

- Secured **Global Rank under 8K** in Leetcode Weekly Contest(464).
- Achieved **National Rank 52** in HCL Amplified Hackathon.
- Qualified GATE 2025, demonstrating strong knowledge in core Computer Science subjects including Operating Systems, DBMS, Computer Networks, and Data Structures and Algorithms.